Upper Bounds for High-Utility Episode Mining under a Maximum Duration Constraint: Problem Analysis, Verified Baseline and Research Plan

Working draft — not a manuscript; contains no experimental results

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Status of this document

This draft replaces an earlier version (repository commit `037d41f61`) that is withdrawn. The earlier version (i) reported speedups, memory reductions and pruning ratios for which no implementation or experiment existed; (ii) described a hardware platform that was not used; (iii) stated that the authors' code of UMEpi was used, although no such code was found; (iv) misstated the upper bound of HUE-Span (it summed utilities to the end of the sequence, whereas Definition 16 of [4] stops at $t_s + maxDur - 1$), so that its running example did not show what it claimed; (v) proposed a bound (RDUB) that is not valid for the episodes mined by HUE-Span (Section 4); (vi) contained four references whose DOIs point to other works and two references that could not be found (Section 6). Everything below is either taken from a cited source that was read, verified by a stated procedure, or explicitly marked as open.

1 Problem definition

We adopt the redefined problem of HUE-Span [4] without change. A complex event sequence is $CES = \langle (tSE_{t_1}, t_1), \ldots, (tSE_{t_n}, t_n) \rangle$ with $t_1 < \cdots < t_n$, where each tSE_{t_i} is a set of events. Each event e has an external utility $p(e) > 0$ and, at time t_i, an internal utility $q(e, t_i) > 0$; $u(e, t_i) = p(e)\, q(e, t_i)$ and $u(SE, t_i) = \sum_{e \in SE} u(e, t_i)$.

Definition 1 (Episode, occurrence, minimal occurrence [4, Defs. 2–4])**.** *An episode is $\alpha = \langle SE_1, \ldots, SE_k \rangle$ with non-empty event sets SE_i. An interval $[t_s, t_e]$ is an occurrence of α if α occurs in it with SE_1 at t_s and SE_k at t_e; it is minimal if no other occurrence of α is a sub-interval of it. $moSet(\alpha)$ denotes the set of minimal occurrences.*

Definition 2 (Redefined utility [4, Def. 12])**.** *For a minimal occurrence $[t_s, t_e]$ of $\alpha = \langle SE_1, \ldots, SE_k \rangle$, $u(\alpha, [t_s, t_e]) = u(SE_1, t_s) + \sum_{i=2}^{k-1} \max_x u(SE_i, t_{g(i)_x}) + u(SE_k, t_e)$, where the maximum ranges over the time points at which SE_i can be matched in the occurrence; $u(\alpha) = \sum_{mo \in moSet(\alpha)} u(\alpha, mo)$.*

Definition 3 (Maximum duration and problem [4, Defs. 13–14])**.** *A minimal occurrence satisfies the maximum duration if $t_e - t_s + 1 \leq maxDur$ (the printed Definition 13 reads $t_e - t_e + 1$, an evident typo; the reference implementation uses $t_s + maxDur - 1$ as the last admissible time point). Given minUtil and maxDur, the task is to find every episode with $u(\alpha) \geq minUtil$, counting only minimal occurrences that satisfy maxDur. In [4] and in its implementation, minUtil is a fraction of the total utility of the sequence.*

[Open: Definition 12 does not state how the candidate time points $t_{g(i)_x}$ of the middle event sets are restricted (e.g. whether the choices for different i must respect the order of the episode). The reference implementation computes the maximum with a finite-state procedure (`ComplexSequence.getMaximalUtility`) and only during serial extensions. A brute-force oracle following the text of Definition 12 is needed before the reference implementation can be called correct with respect to the paper (Section 5).]

2 Baselines that solve the same problem

Table 1: Candidate baselines and their status (checked on 20 October 2026).

Algorithm	Problem solved	Status
HUE-Span [4]	this problem (redefined utility)	Authors' Java code in SPMF [6]. Ported to C in the `dm` framework (`hue_span`). On the running example of [4] (minUtil = 0.3, maxDur = 3) the port and the Java code return the same 66 episodes with the same utilities and the same six counters printed by `printStats`; a randomized parity test over six flag configurations is under way.
HUE-Span variants	same	ERU only, EEUCP only, EWU: obtained from the same code by its flags, as in [4, Fig. 3].
UP-Span [2]	traditional utility (no maximum)	In SPMF. Comparable only for the traditional problem; [4] reports that it misses up to 65% of the episodes of the redefined problem.
UMEpi [5]	a different definition (no maximum over time points; duration $T_e - T_s \leq MTD$)	No public code found. The copy in the repository is the arXiv preprint 1912.11670v1, not the published version; the computation of $moSet(\beta)$ after a concatenation and the choice of time points within a minimal occurrence are not specified. Not usable as a baseline for this problem.
TSpan [3]	traditional utility	No public code; [4] describes it as appearing incomplete.

3 Existing upper bounds

For a minimal occurrence $[t_s, t_e]$ of $\alpha = \langle SE_1, \ldots, SE_k \rangle$ satisfying $maxDur$, with $tu(t)$ the total utility of the events at t:

$$\mathrm{EWU}(\alpha, [t_s, t_e]) = \sum_{i=1}^{k-1} u(SE_i, t_{g(i)}) + \sum_{j=t_e}^{t_s+maxDur-1} tu(j) \qquad \text{[4, Def. 16]} \quad (1)$$

$$\mathrm{ERU}(\alpha, [t_s, t_e]) = \sum_{i=1}^{k} u(SE_i, t_{g(i)}) + u(rSE_{t_e}, t_e) + \sum_{j=t_e+1}^{t_s+maxDur-1} tu(j) \qquad \text{[4, Def. 18]} \quad (2)$$

where $u(rSE_{t_e}, t_e)$ is the utility of the events at t_e that follow the last event of SE_k in the processing order. Both bounds are already restricted to the duration horizon $t_s + maxDur - 1$; summed over $moSet(\alpha)$ they bound the utility of every super-episode obtained by concatenations [4, Lemma 1]. The reference implementation computes ERU exactly in this form (utility of the occurrence, `getIrutil` at t_e, and the total utility of $[t_e + 1, t_s + maxDur - 1]$).

Example 1 (Correction of the earlier running example)**.** *In the sequence of the earlier draft (A at $t = 1, 4, 7$ with utilities 10, 5, 15; B at $t = 2, 4, 6, 8$; $maxDur = 4$; $\mu = 35$), the episode*

$\gamma = \langle A, B, B\rangle$ *has the single valid minimal occurrence* $[1, 4]$ *(A at 1, B at 2 and 4). Definition 16 gives* $\mathrm{EWU}(\gamma) = u(A, 1) + u(B, 2) + tu(4) = 10 + 3 + 11 = 24 < 35$*, so HUE-Span also prunes* γ*. The earlier value 101 summed* $tu(5), \ldots, tu(10)$*, which lie beyond the horizon* $1 + 4 - 1 = 4$ *and are not part of EWU.*

4 The proposed bound RDUB is not valid for this problem

The earlier draft replaced, for each future time point in the horizon, the total utility $tu(t)$ by the largest single-event utility $\max_e u(e, t)$. This is not an upper bound when an extension may contain several events at the same time point, which the episodes of HUE-Span allow (simultaneous concatenation).

Proposition 1. *The bound* $\mathrm{RDUB}(\alpha) = \sum_{mo}[u(\alpha, mo) + \sum_{t\in(t_e,\, t_s+maxDur-1]} \max_e u(e, t)]$ *can be smaller than the utility of a super-episode of* α*.*

Proof. Let $CES = \langle(\{A\}, 1), (\{B, C\}, 2)\rangle$ with $u(A, 1) = 1$, $u(B, 2) = u(C, 2) = 5$, $maxDur = 2$ and $\mu = 10$. Then $\mathrm{RDUB}(\langle A\rangle) = 1 + 5 = 6 < \mu$, so $\langle A\rangle$ and its super-episodes would be pruned, but $\langle(A), (BC)\rangle$ has the minimal occurrence $[1, 2]$ and utility $1 + 10 = 11 \geq \mu$. □

For serial episodes with one event per position, the per-time-point maximum is a plausible bound, but that is a different (restricted) problem, for which HUE-Span is not the natural baseline, and the earlier proof relied on the unproven claim that every minimal occurrence of an extension extends a minimal occurrence of the prefix with the same prefix utility. [Open: whether any tightening of ERU remains valid for complex episodes, and by how much it can reduce the search; the slack of ERU (ERU minus the utility actually reached by extensions) has not been measured.]

5 Research plan

1. **Oracle.** Brute-force enumeration of episodes and minimal occurrences on small random sequences, with the utility of Definition 12 computed by exhaustive search over the admissible time points; compare with the HUE-Span reference implementation. Any difference must be reported with a minimal counterexample.

2. **Parity.** Complete the randomized parity test of the C port against the Java code (all six flag configurations).

3. **Slack measurement.** Instrument the port to record, per expanded episode, ERU and the largest utility reached in its subtree, to determine whether a tighter bound can matter before designing one.

4. **Fair comparison.** Any new algorithm is compared with the C port of HUE-Span in the same framework (same language, same input parsing), on the datasets of [4] (Retail, Kosarak, T25I10N1KD10KQ10F5 with synthetic utilities); the generation procedure and seeds of the utilities must be documented. Platform: Intel Core i7-10700, 32 GiB RAM, Ubuntu 24.04, GCC 13.3, single thread.

6 Corrections to the reference list

Each DOI of the earlier list was resolved with the Crossref API on 20 October 2026. Removed: "Mining closed episodes with overlap" (Tatti and Cule, TKDD 2012) and "Mining episodes with a threshold on duration" (Cule, Goethals and Tatti, ICDM 2009): their DOIs resolve to unrelated papers and no record with these titles was found in OpenAlex. Corrected: USpan

(DOI 10.1145/2339530.2339636), FHM (DOI 10.1007/978-3-319-08326-1_9, pp. 83–92), HUE-Span (pp. 169–184), MaxFEM (pp. 86–98), TKE (pp. 832–845), Wu et al. 2017 (pp. 215–229).

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